

Machine Learning Engineer & Researcher with 7+ years of experience in AI-driven decision-making, ranking, and search. Proven track record in scalable ML systems and publishing cutting-edge reinforcement learning and large language model research.

Experience

- Mila - Quebec AI Institute** — *ML Research Assistant* Montreal, Canada — Dec 2023 – Present
 - Researching scalable deep RL architectures, and methods for steering LLM-contributing to AI alignment and interpretability.
 - Currently exploring video models for world modeling and model-based reinforcement learning.
- Mila - Quebec AI Institute** — *MSc Research Student* Montreal, Canada — Jan 2021 – Dec 2023
 - Researched model-based reinforcement learning and various forms of adaptation in continual learning, publishing in top AI conferences.
- CafeBazaar Inc.** — *Data Scientist* Tehran, Iran — Sep. 2017 – Jan. 2021
 - CafeBazaar is Iran’s largest third-party Android marketplace, **serving 40M+ users**.
 - Personalization Team:
 - Developed and deployed an app recommender system, improving **install rates from 1.6% to 2.5%**.
 - Designed a diversification strategy to broaden recommended app categories, enhancing user engagement.
 - Search-ads Team:
 - As a core founding member, led data science, mechanism design, and optimization for the search advertising system. ([Technical Report](#))
 - Helped establish Search-Ads as a top 3 daily revenue driver for the company.
 - Search Team:
 - Improved the app ranking system and built NLP-based services like spell-checking and query suggestion.
 - Reduced the **search bounce rate from 41% to 35%**, increasing search satisfaction.
- Chair for CAMPAR** — *Research Assistant* Munich, Germany — Jul. 2019 – Oct. 2019
 - Worked on 3D camera re-localization in dynamic environments. Developed a valid pose generator using rendered scenes for 3RScan data.

Education

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| <ul style="list-style-type: none">University of Montreal - Mila
M.Sc. in Computer Science (Artificial Intelligence) | <ul style="list-style-type: none">Sharif University of Technology
B.Sc. in Computer Science |
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Publications * DENOTES EQUAL CONTRIBUTION.

- Steering Large Language Model Activations in Sparse Spaces**
Under Review ([paper](#))
Rahimi-Kalahroudi, A.*, Bayat, R.*, Pezeshki, M., Chandar, S., Vincent, P.
- Partial Models for Building Adaptive Model-Based Reinforcement Learning Agents**
Conference on Lifelong Learning Agents, **CoLLAs 2024** ([paper](#))
Alver, S., **Rahimi-Kalahroudi, A.**, Precup, D.
- Replay Buffer with Local Forgetting for Adapting to Local Environment Changes in Deep MBRL**
Conference on Lifelong Learning Agents, **CoLLAs 2023** (*Oral*) ([paper](#))
and **NeurIPS 2022** - Deep Reinforcement Learning Workshop
Rahimi-Kalahroudi, A., Rajendran, J., Momennejad, I., Van Seijen, H., Chandar, S.
- Towards Evaluating Adaptivity of Model-Based Reinforcement Learning Methods**
International Conference on Machine Learning, **ICML 2022** ([paper](#), [code](#))
Rahimi-Kalahroudi, A.*, Wan, Y.*, Rajendran, J., Momennejad, I., Chandar, S., Van Seijen, H.

Honors and Awards

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| <ul style="list-style-type: none">Best M.Sc. Thesis - Canadian Artificial Intelligence Association (CAIAC)Exceptional Grade for the M.Sc. Thesis (Top 1%) - University of MontrealBourse D’excellence DIRO - University of MontrealM.Sc. Tuition Exemption Scholarships (Bourse C) - University of MontrealInternational Undergraduate Excellence Award - Technical University of MunichSilver Medal - 24th Iranian National Olympiad in Informatics (INOI) | <ul style="list-style-type: none">Guelph, Canada — 2024Montreal, Canada — 2023Montreal, Canada — 2022Montreal, Canada — 2021Munich, Germany — 2019Tehran, Iran — 2014 |
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Skills

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| <ul style="list-style-type: none">Programming: Python, Scala, and C++.ML Tools: PyTorch, Tensorflow, NumPy, and JAX. | <ul style="list-style-type: none">Data Technologies: Apache Spark, Hadoop, Airflow, and Kafka.Misc: Docker, Apptainer, Bash, Slurm, and $\LaTeX$ |
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